

Mathematical analysis of a secretion model.

Elías Vera-Sigüenza¹ · Martin Golubitsky² ·
James Sneyd¹

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Abstract We demonstrate the mathematical basis for which a previous salivary gland acinar cell secretion model, constructed by Sigüenza et al. (2017), supported experimental results by Pena-Munzenmayer et al. (2015). The latter, revealed for the first time, the presence of a monovalent cation-dependent anion exchanger encoded by the Slc4a9 (Ae4) gene in these cells. The mechanism, previously thought to be expressed exclusively in the plasma membrane (PM) of salivary gland duct cells, supports the accumulation of intracellular chloride (Cl^-) and the regulation of intracellular pH (Pena-Munzenmayer et al., 2016). Through the use of numerical continuation techniques we demonstrate: 1) That the exchanger's dependence on monovalent cations renders the Ae4 as a paramount mechanism for supporting optimal fluid flow in acinar cells; 2) we also demonstrate that Sigüenza et al. (2017) model's choice is independent to the results; and 3) we investigate the sensitivity of the model to various parameters, we conclude that the models results are independent to the choice of parameters when these vary in a narrow physiologically sensible range.

Keywords Acinar Cell, Ae4, Ae2, Exchangers, Salivary Gland, Mathematical Modelling

¹Department of Mathematics, The University of Auckland, Level 2, Building 303, 38 Princes Street, Auckland CBD, New Zealand, E-mail: esig526@aucklanduni.ac.nz

Department of Mathematics, The Ohio State University, 618 Math Tower, 231 W 18th Avenue, Columbus OH, United States of America

1 Introduction

The production of salivary fluid occurs in two stages. The first of these consists in the creation of a fluid referred to as “primary saliva” (Martinez et al, 1966). This secretion is the result of transepithelial water movement across salivary gland acinar cells. It is characterised by its elevated chloride (Cl^-) and sodium (Na^+) content and a relatively low concentration of potassium (K^+). The ion channels and transporters, expressed in the plasma membrane (PM) of the acini, promote vectorial Cl^- transport in the secretory direction (from the serosal to the luminal side). The transcellular movement of Cl^- is directly responsible for creating an osmotic gradient that water follows (Silva et al, 1977). The basolateral $\text{Na}^+/\text{K}^+/\text{2Cl}^-$ cotransporter, encoded by the *Slc12a2* (*Nkcc1*) gene, is thought to be the main molecular mechanism in charge of accumulating the intracellular space with Cl^- above its equilibrium potential (Melvin et al, 2005). Studies have demonstrated that lack of *Nkcc1* expression in the PM of acinar cells results in a severely impaired rate flow, of about 70% less than those cells that expressed the mechanism. The residual fluid secretion is bicarbonate (HCO_3^-) dependent (Evans et al, 2000).

For many years it was hypothesised that the basolateral $\text{HCO}_3^-/\text{Cl}^-$ exchanger, encoded by the *Slc4a2* (*Ae2*) gene, was directly responsible for the HCO_3^- -mediated salivation in acinar cells (Evans et al, 2000; Nguyen et al, 2004). The *Ae2* can be found in nearly all mammalian epithelia, as it is in charge of regulating intracellular pH and transepithelial acid-base balance in the cell (Holappa et al, 2001). Much to researchers’ surprise, recent experiments by Peña-Münzenmayer et al (2015) revealed that not only the *Ae2* is not involved in supporting the *Nkcc1* cotransporter in the accumulation of intracellular Cl^- , but that lack of *Ae2* expression produced no changes to the salivary gland fluid flow rate. Instead, they found that another HCO_3^- -dependent Cl^- transporting protein was responsible for the remaining HCO_3^- -mediated salivation. The mechanism identified is a monovalent cation-dependent $\text{HCO}_3^-/\text{Cl}^-$ exchanger encoded by the *Slc4a9* (*Ae4*) gene. This exchanger was previously known only to be expressed in the basolateral PM of salivary gland duct cells. The experimental results show that systemic *Ae4* knock-out mice exhibited a 30% reduction in stimulated salivary flow (Peña-Münzenmayer et al, 2015).

Sigüenza et al. (2017) devised a dynamical model of a single salivary gland acinar cell that includes these $\text{HCO}_3^-/\text{Cl}^-$ exchanger mechanisms with the aim of understanding the complex behaviour behind Peña-Münzenmayer et al (2015) results. Through ‘in-silico’ knock-out simulations of the *Ae2* and *Ae4*, the model’s results supported Peña-Münzenmayer et al (2016) findings and suggested that the observed behaviour is due to the monovalent-cation dependence of the *Ae4*. Here we analyse the mathematical model and the results as presented by Sigüenza et al. (2017). Through the use of mathematical continuation techniques we explore the mathematical basis for the model’s predictions and investigate its sensitivity to a set of particular parameters relevant to fluid flow.

2 Secretion Model

Sigüenza et al. (2017) constructed a well-stirred dynamical secretory model of a single salivary gland acinar cell based on the conservation of ions across three compartments; the interstitium (serosal media), intracellular, and the acinar lumen. The interstitial compartment assumes a constant concentration of ions at all times and thus it renders the model equivalent as placing a single cell in an infinite bath of ions. The concentration of a particular ionic species in the intracellular and luminal media varies according to fluxes (J) determined by a Na^+ - K^+ - 2Cl^- - co-transporter (Nkcc1), a Na^+ / K^+ -ATPase (NaK), a $\text{Cl}^-/\text{HCO}_3^-$ exchanger (Ae2), a monovalent cation-dependent $\text{Cl}^-/\text{HCO}_3^-$ exchanger (Ae4), a H^+/Na^+ antiporter (Nhe1), a CO_2 membrane flux, and an intracellular HCO_3^- buffer. Water secretion in the model is driven by an increase in free cytosolic $[\text{Ca}^{2+}]_i$. This is a parameter of the model and it takes on two distinct values: $[\text{Ca}^{2+}]_i = 0.58 \text{ mM}$ and $[\text{Ca}^{2+}]_i = 0.2 \text{ } \mu\text{M}$, these represent the resting and simulated secretion states, respectively. Ca^{2+} activates Cl^- channels (CaCC) in the apical PM portion of the cell providing a route for Cl^- to exit into the lumen. It also serves to activate basolateral K^+ channels (CaKC), causing the efflux of K^+ out of the cell into the interstitium. This efflux maintains a cell membrane potential that supports the extrusion of Cl^- into the acinar lumen. In addition to the Ca^{2+} -activated currents, the model also includes two paracellular fluxes: Na^+ and K^+ . These are driven by the transepithelial negative potential established by Cl^- efflux into the lumen. In this way, the ionic species rates of change with respect to time in the cell and the luminal compartments are defined as a set of ordinary differential equations,

$$\begin{aligned}
\frac{d([\text{Cl}^-]_i \omega_i)}{dt} &= 2J_{\text{Nkcc1}} + J_{\text{Ae4}} + J_{\text{Ae2}} - J_{\text{CaCC}}, \\
\frac{d([\text{Na}^+]_i \omega_i)}{dt} &= J_{\text{Nkcc1}} + J_{\text{Nhe1}} - 3J_{\text{NaK}} - J_{\text{Ae4}}, \\
\frac{d([\text{K}^+]_i \omega_i)}{dt} &= J_{\text{Nkcc1}} + 2J_{\text{NaK}} - J_{\text{CaKC}}, \\
\frac{d([\text{H}^+]_i \omega_i)}{dt} &= J_{\text{Buffer}} - J_{\text{Nhe1}}, \\
\frac{d([\text{CO}_2]_i \omega_i)}{dt} &= J_{\text{CO}_2} - J_{\text{Buffer}}, \\
\frac{d([\text{HCO}_3^-]_i \omega_i)}{dt} &= J_{\text{Buffer}} - 2J_{\text{Ae4}} - J_{\text{Ae2}}, \\
\omega_l \frac{d[\text{Na}^+]_l}{dt} &= J_{\text{Na}}^t - q_{\text{tot}}[\text{Na}^+]_l, \\
\omega_l \frac{d[\text{K}^+]_l}{dt} &= J_{\text{K}}^t - q_{\text{tot}}[\text{K}^+]_l, \\
\omega_l \frac{d[\text{Cl}^-]_l}{dt} &= J_{\text{CaCC}} - q_{\text{tot}}[\text{Cl}^-]_l.
\end{aligned}$$

Here ω_l represents the acinar lumen volume, it is assumed a constant of the model. The variable ω_i represents the volume of the cell which is given by the

difference between the apical and basolateral fluid flow, q_a and q_b , respectively:

$$\frac{d\omega_i}{dt} = q_b - q_a,$$

where q_a and q_b are given by:

$$\begin{aligned} q_a &= RT L_a \left([\text{Cl}^-]_l + [\text{Na}^+]_l + [\text{K}^+]_l + \frac{x_l}{\omega_l} \right. \\ &\quad \left. - \left([\text{Cl}^-]_i + [\text{Na}^+]_i + [\text{K}^+]_i + [\text{HCO}_3^-]_i + [\text{H}^+]_i + [\text{CO}_2]_i + \frac{x_i}{\omega_i} \right) \right), \\ q_b &= RT L_b \left([\text{Cl}^-]_i + [\text{Na}^+]_i + [\text{K}^+]_i + [\text{HCO}_3^-]_i + [\text{H}^+]_i + [\text{CO}_2]_i \right. \\ &\quad \left. + \frac{x_i}{\omega_i} - \left([\text{Cl}^-]_e + [\text{Na}^+]_e + [\text{K}^+]_e \right) \right), \end{aligned}$$

here R is the universal gas constant and T is the temperature in Kelvin. The parameters x_i and x_l denote the moles of large negatively charged molecules (with valence $z_x \leq -1$) that are trapped inside the cell and in the lumen. L_a and L_b represent the apical and basolateral PM permeabilities to water. The model includes a paracellular fluid transport (through the tight junctions), this is given by:

$$q_t = RT L_t \left([\text{Cl}^-]_l + [\text{Na}^+]_l + [\text{K}^+]_l + \frac{x_l}{\omega_l} - \left([\text{Cl}^-]_e + [\text{Na}^+]_e + [\text{K}^+]_e \right) \right),$$

where L_t is the permeability to water of the tight junction. The total fluid flow rate of the model is given by:

$$q_{\text{tot}} = q_a + q_t.$$

The expression above represents the total influx of water into the lumen ($q_a + q_t$). However, this is matched by a water efflux into the salivary duct (out of the lumen). Thus q_{tot} also represents the fluid flow rate out of the lumen, a consequence of assuming a fixed luminal volume (ω_l) (Sigüenza et al., 2017).

Finally, the presence of two different membrane potentials on different portions of the same cellular membrane (apical and the basolateral) result in two coupled membrane potential equations modelled according to Kirchhoff's current law for a simple resistor-capacitor circuit:

$$\begin{aligned} C_m \frac{dV_b}{dt} &= -I_{\text{CaKC}} - I_{\text{NaK}} + (I_{\text{K}}^t + I_{\text{Na}}^t), \\ C_m \frac{dV_a}{dt} &= -I_{\text{CaCC}} - (I_{\text{K}}^t + I_{\text{Na}}^t), \end{aligned}$$

In the equations above the term I_{CaCC} , represents the Ca^{2+} -activated Cl^- channel current in the apical membrane. The terms I_{CaKC} and I_{NaK} represent the current due to Ca^{2+} -activated K^+ channels and NaK-ATPase pumps in the basolateral plasma membrane. The terms I_{K}^t and I_{Na}^t represent the Na^+ and K^+

currents through the tight junctions. The parameter C_m is the capacitance of the cell membrane. In this model the apical membrane potential is negative as it is measured from the lumen to the cytoplasm. The basolateral membrane potential, on the other hand, is measured from the interstitium to the cytoplasm. To see full details of the model, refer to Sigüenza et al., (2017).

3 Simplified Model

To begin our analysis, we carefully seek to reduce the system of equations with the aim of simplifying the task at hand. In their paper Sigüenza et al. (2017), suggest that the reason behind the observed decrease in salivary flow upon Ae4 deletion is its monovalent cation dependence, particularly Na^+ in their model. Because the main ions present in primary saliva are Cl^- , K^+ , and Na^+ , it follows that if the model suggestions are correct, the system's main dynamics could be conveyed by the equations describing the rates of change of these ions. Then, to simplify the system we assume the following ionic concentrations to be constants of the model: $[\text{Ca}^{2+}]_i$, $[\text{HCO}_3^-]_i$, $[\text{CO}_2]_i$, $[\text{H}^+]_i$, $[\text{K}^+]_l$, $[\text{Cl}^-]_l$, and $[\text{Na}^+]_l$. Additionally, we assume the cellular and luminal volumes to be constant along with the apical and basolateral membrane potentials (ω_i , ω_l , V_a , and V_b). We have chosen to focus on the stimulated steady states of the cell, as it is here where the distinction between Ae4 knockout and non-knockout fluid secretion is observable. Under these assumptions, we were able to reduce the system of equations to a 3 equation system,

$$\omega_i \frac{d[\text{Na}^+]_i}{dt} = J_{\text{Nkcc1}} - 3J_{\text{NaK}} + J_{\text{Nhe1}} - \textcolor{red}{G}_4 \tilde{J}_{\text{Ae4}}, \quad (1)$$

$$\omega_i \frac{d[\text{K}^+]_i}{dt} = J_{\text{Nkcc1}} + 2J_{\text{NaK}} - \frac{I_{\text{CaKC}}}{Fz_{\text{K}}}, \quad (2)$$

$$\omega_i \frac{d[\text{Cl}^-]_i}{dt} = 2J_{\text{Nkcc1}} + \textcolor{red}{G}_4 \tilde{J}_{\text{Ae4}} + \textcolor{red}{G}_2 \tilde{J}_{\text{Ae2}} - \frac{I_{\text{CaCC}}}{Fz_{\text{Cl}}}, \quad (3)$$

where z^{Cl} and z^{K} are the respective ion valences, and $J_{\text{Aej}} = \textcolor{red}{G}_j \tilde{J}_{\text{Aej}}$, for $j = 2$ or 4 . To further simplify the analysis of the system's variables, we non-dimensionalise equations (1-3). Through a simple dimensional analysis we see that the dimensional units corresponding to the left hand side are,

$$\left[\omega_i \frac{d[X]_i}{dt} \right] = \mu m^3 \left(\frac{\text{mM}}{\text{second}} \right) = \mu m^3 \left(\frac{\text{amol}}{\text{second } \mu m^3} \right) = \frac{\text{amol}}{\text{second}}.$$

and thus the units corresponding to the right hand side must be,

$$\left[\sum_i J_i \right] = \frac{\text{amol}}{\text{second}}.$$

Accordingly, we seek a set of parameters that are sensible to the analysis of our system. The following, were chosen for simplicity: G_{Nhe1} with units of (amol/second),

q_{a_0} with units of ($\mu m^3/\text{second}$), ω_l with units of (μm^3), and T in seconds. If we multiply the left hand side of the equations by,

$$\left[\frac{q_{a_0} T}{G_{\text{Nhe1}} \omega_l} \right]^{-1} = \left(\frac{\mu m^3 \text{ second}}{\mu m^3 \text{ amol}} \right)^{-1} = \left(\frac{\text{second}}{\text{amol}} \right)^{-1}$$

we obtain:

$$\begin{aligned} \omega_i \frac{d[\text{Na}^+]_i}{dt} &= \left(\frac{\omega_l \omega G_{\text{Nhe1}}}{q_{a_0} T} \right) \frac{dx}{d\tau} = \alpha \frac{dy}{d\tau}, \\ \omega_i \frac{d[\text{K}^+]_i}{dt} &= \left(\frac{\omega_l \omega G_{\text{Nhe1}}}{q_{a_0} T} \right) \frac{dy}{d\tau} = \alpha \frac{dx}{d\tau}, \\ \omega_i \frac{d[\text{Cl}^-]_i}{dt} &= \left(\frac{\omega_l \omega G_{\text{Nhe1}}}{q_{a_0} T} \right) \frac{dz}{d\tau} = \alpha \frac{dz}{d\tau}, \end{aligned}$$

where, $\alpha = \left(\frac{\omega_l \omega G_{\text{Nhe1}}}{q_{a_0} T} \right)$. x is the non-dimensional $[\text{Na}^+]_i$, y is the non-dimensional $[\text{K}^+]_i$, and z is the non-dimensional $[\text{Cl}^-]_i$. Accordingly, the right hand side of the equations is given by,

$$\sum_i J_i = \sum_i G_{\text{Nhe1}} \hat{J}_i.$$

Now, $\hat{J}_i$ is the non-dimensional “flux” term for the i^{th} transporter. Let $\epsilon = \frac{G_{\text{Nhe1}}}{\alpha}$, then the non-dimensional model is given by

$$\begin{aligned} \frac{dx}{d\tau} &= \epsilon \left(\hat{J}_{\text{Nkcc1}} - 3\hat{J}_{\text{NaK}} + \hat{J}_{\text{Nhe1}} - \textcolor{red}{G}_4 \hat{J}_{\text{Ae4}} \right), \\ \frac{dy}{d\tau} &= \epsilon \left(\hat{J}_{\text{Nkcc1}} + 2\hat{J}_{\text{NaK}} - \hat{J}_{\text{K}}, \right. \\ \frac{dz}{d\tau} &= \epsilon \left(2\hat{J}_{\text{Nkcc1}} + \textcolor{red}{G}_4 \hat{J}_{\text{Ae4}} + \textcolor{red}{G}_2 \hat{J}_{\text{Ae2}} + \hat{J}_{\text{Cl}} \right). \end{aligned}$$

Expanding the equations above we obtain the full non-dimensional system (see Appendix section):

$$\frac{dx}{d\tau} = \frac{\lambda_1 - \lambda_2(xy z^2)}{\lambda_3 + \lambda_4(xy z^2)} - 3 \frac{\lambda_5 x^3}{\lambda_6 + \lambda_7 x^3} + \left(\lambda_8 - \frac{\lambda_9 x}{x + \lambda_{10}} \right) - \textcolor{red}{G}_4 (\lambda_{11} x - \lambda_{12} z), \quad (4)$$

$$\frac{dy}{d\tau} = \frac{\lambda_1 - \lambda_2(xy z^2)}{\lambda_3 + \lambda_4(xy z^2)} + 2 \frac{\lambda_5 x^3}{\lambda_6 + \lambda_7 x^3} - (\lambda_{13} - \lambda_{14} \ln(y)), \quad (5)$$

$$\frac{dz}{d\tau} = 2 \frac{\lambda_1 - \lambda_2(xy z^2)}{\lambda_3 + \lambda_4(xy z^2)} + \textcolor{red}{G}_4 (\lambda_{11} x - \lambda_{12} z) + (\lambda_{15} - \lambda_{16} \ln(z)) + \textcolor{red}{G}_2 \left(\lambda_{17} - \frac{\lambda_{18} z}{z + \lambda_{19}} \right), \quad (6)$$

In this way, the total non-dimensional cellular fluid rate, represented by q_{tot} in the full model, is given by:

$$Q = \lambda_{20} - \lambda_{21} \left(x + y \right). \quad (7)$$

The parameters of the system, λ_i ($i = 1, 2, \dots, 21$), can be found in Table 1.

Table 1: Parameters of the model.

Parameter	Value	Parameter	Value
λ_1	16.266	λ_{14}	71.750
λ_2	2074819.479	λ_{15}	191.363
λ_3	1.03	λ_{16}	56.925
λ_4	1385200	λ_{17}	20.421
λ_5	38.545	λ_{18}	16.179
λ_6	0.00002809	λ_{19}	5.6
λ_7	0.641	λ_{20}	0.472
λ_8	8.466	λ_{21}	0.003
λ_9	3.473	G_4	6.74e-5
λ_{10}	15	G_2	0.347
λ_{11}	4589.648	x_0	12.29
λ_{12}	13.423	y_0	89.19
λ_{13}	-313.123	z_0	37.13

Finally, because we are only interested in the steady states of the system we can further simplify the equations such that only f_1 contains the Ae4 flux term and only f_3 contains the Ae2 flux term. Adding both equations we obtain:

$$f_1 = \frac{\lambda_1 - \lambda_2 s^4 (xyz^2)}{\lambda_3 + \lambda_4 (xyz^2)} - 3 \frac{\lambda_5 s^3 x^3}{\lambda_6 + \lambda_7 s^3 x^3} + \left(\lambda_8 - \frac{\lambda_9 x}{x + \lambda_{10}} \right) - G_4 (\lambda_{11} x - \lambda_{12} z) = 0, \quad (8)$$

$$f_2 = \frac{\lambda_1 - \lambda_2 (xyz^2)}{\lambda_3 + \lambda_4 (xyz^2)} + 2 \frac{\lambda_5 x^3}{\lambda_6 + \lambda_7 x^3} - (\lambda_{13} - \lambda_{14} \ln(y)) = 0, \quad (9)$$

$$f_3 = 3 \frac{\lambda_1 - \lambda_2 s^4 (xyz^2)}{\lambda_3 + \lambda_4 (xyz^2)} - 3 \frac{\lambda_5 s^3 x^3}{\lambda_6 + \lambda_7 s^3 x^3} + \left(\lambda_8 - \frac{\lambda_9 x}{x + \lambda_{10}} \right) + G_2 \left(\lambda_{17} - \frac{\lambda_{18} z}{z + \lambda_{19}} \right) + (\lambda_{15} - \lambda_{16} \ln(z)) = 0. \quad (10)$$

The steady state values of the system can be found in Table 1.

4 Results

To understand the system's dependence on the parameters G_4 and G_2 we turn to mathematical continuation. Let equation's (5)-(7) be represented by:

$$\bar{\mathbf{x}} = \begin{pmatrix} x = x(\tau) \\ y = y(\tau) \\ z = z(\tau) \end{pmatrix}$$

Such that

$$\bar{\mathbf{F}}(\bar{\mathbf{x}}_0; \mathbf{G}_2, \mathbf{G}_4) = \begin{pmatrix} f_1(\bar{\mathbf{x}}_0; \mathbf{G}_2, \mathbf{G}_4) \\ f_2(\bar{\mathbf{x}}_0; \mathbf{G}_2, \mathbf{G}_4) \\ f_3(\bar{\mathbf{x}}_0; \mathbf{G}_2, \mathbf{G}_4) \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix}.$$

Using the implicit function theorem, we let

$$x = x(\mathbf{G}_i), \quad y = y(\mathbf{G}_i), \quad z = z(\mathbf{G}_i);$$

and seek to find a solution $\bar{\mathbf{x}}(\mathbf{G}_i)$ for

$$\frac{d\bar{\mathbf{F}}}{d\mathbf{G}_i} = \bar{\mathbf{0}},$$

where the subscript i refers to either 4 or 2. Finally, it should also be noted that the physiological range of the parameter $\mathbf{G}_2$ is $[0, 0.347]$. Similarly, the physiological range of the parameter $\mathbf{G}_4$ is $[0, 6.74e-5]$. Values outside this range are outside the scope of our interest. We solved this by using the numerical continuation MATLAB package, MATCONT. We are interested in understanding why the Ae4 flux term has a significant impact in the dynamics of the model whereas the Ae2 does not. To do so we continue our investigation by computing

$$\frac{dQ}{d\mathbf{G}_i} = -\lambda_{21} \left(\frac{dx}{d\mathbf{G}_i} + \frac{dy}{d\mathbf{G}_i} \right). \quad (11)$$

So we calculate

$$\frac{d\bar{\mathbf{F}}}{d\mathbf{G}_i} = \frac{\partial \bar{\mathbf{F}}}{\partial \bar{\mathbf{x}}} \frac{d\bar{\mathbf{x}}}{d\mathbf{G}_i} + \frac{\partial \bar{\mathbf{F}}}{\partial \mathbf{G}_i} = \bar{\mathbf{0}}, \quad (12)$$

where i is 2 or 4, accordingly.

$$\frac{\partial \bar{\mathbf{F}}}{\partial \mathbf{G}_4} = \begin{pmatrix} z\lambda_{12} - x\lambda_{11} \\ 0 \\ 0 \end{pmatrix}, \quad \frac{\partial \bar{\mathbf{F}}}{\partial \mathbf{G}_2} = \begin{pmatrix} 0 \\ 0 \\ \lambda_{17} - \frac{\lambda_{18}z}{\lambda_{19}+z} \end{pmatrix}.$$

Now we solve,

$$\begin{pmatrix} \frac{\partial f_1}{\partial x} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial x} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \\ \frac{\partial f_3}{\partial x} & \frac{\partial f_3}{\partial y} & \frac{\partial f_3}{\partial z} \end{pmatrix} \begin{pmatrix} \frac{dx}{d\mathbf{G}_i} \\ \frac{dy}{d\mathbf{G}_i} \\ \frac{dz}{d\mathbf{G}_i} \end{pmatrix} = - \begin{pmatrix} \frac{\partial f_1}{\partial \mathbf{G}_i} \\ \frac{\partial f_2}{\partial \mathbf{G}_i} \\ \frac{\partial f_3}{\partial \mathbf{G}_i} \end{pmatrix},$$

We use Cramer's Rule to investigate the detrimants that constitute the values

$$\frac{dx}{d\mathbf{G}_i} = \frac{\text{Det}(J_x)}{\text{Det}(J)}, \quad \frac{dy}{d\mathbf{G}_i} = \frac{\text{Det}(J_y)}{\text{Det}(J)},$$

where

$$J_x = \begin{pmatrix} -\frac{\partial f_1}{\partial G_i} & \frac{\partial f_1}{\partial y} & \frac{\partial f_1}{\partial z} \\ -\frac{\partial f_2}{\partial G_i} & \frac{\partial f_2}{\partial y} & \frac{\partial f_2}{\partial z} \\ -\frac{\partial f_3}{\partial G_i} & \frac{\partial f_3}{\partial y} & \frac{\partial f_3}{\partial z} \end{pmatrix}, \quad J_y = \begin{pmatrix} \frac{\partial f_1}{\partial x} & -\frac{\partial f_1}{\partial G_i} & \frac{\partial f_1}{\partial z} \\ \frac{\partial f_2}{\partial x} & -\frac{\partial f_2}{\partial G_i} & \frac{\partial f_2}{\partial z} \\ \frac{\partial f_3}{\partial x} & -\frac{\partial f_3}{\partial G_i} & \frac{\partial f_3}{\partial z} \end{pmatrix}.$$

Let

$$\begin{aligned} \beta_1 &= 2x^2y^2z^3(\lambda_2\lambda_3 + \lambda_1\lambda_4)^2, \\ \beta_2 &= -xyz^2(\lambda_2\lambda_3 + \lambda_4(\lambda_1 + (2\lambda_3 + xyz^2\lambda_4)\lambda_{14})) - \lambda_3^2\lambda_{14}, \\ \beta_3 &= \lambda_{12}G_4(\lambda_3 + xyz^2\lambda_4)^2 - 2xyz(\lambda_2\lambda_3 + \lambda_1\lambda_4), \\ \beta_4 &= y(\lambda_3 + xyz^2\lambda_4)^4. \end{aligned}$$

Similarly, let

$$\begin{aligned} \delta_1 &= -6x^2y^2z^2(\lambda_2\lambda_3 + \lambda_1\lambda_4)^2(z + \lambda_{19})^2, \\ \delta_2 &= z(\lambda_3 + xyz^2\lambda_4)^2\beta_2(z + \lambda_{19})^2, \\ \delta_3 &= -\frac{6xyz(\lambda_2\lambda_3 + \lambda_1\lambda_4)}{(\lambda_3 + \lambda_4xyz^2)^2} - \frac{\lambda_{16}}{z} - \frac{\lambda_{18}\lambda_{19}G_2}{(z + \lambda_{19})^2}, \\ \delta_4 &= yz(\lambda_3 + \lambda_4xyz^2)^4(z + \lambda_{19})^2. \end{aligned}$$

Comparing $\text{Det}(J_{x_2})$ and $\text{Det}(J_{x_4})$

$$\begin{aligned} \text{Det}(J_{x_2}) &= \left(\frac{z\lambda_{18}}{z + \lambda_{19}} - \lambda_{17}\right)\left(\frac{\beta_1 - \beta_2\beta_3}{\beta_4}\right), \\ \text{Det}(J_{x_4}) &= (x\lambda_{11} - z\lambda_{12})\left(\frac{\delta_1 + \delta_2\delta_3}{\delta_4}\right). \end{aligned}$$

Substituting the numerical values, we see that the expression

$$\left(\frac{\delta_1 + \delta_2\delta_3}{\delta_4}\right),$$

has a magnitude of 10^0 for all values of $x(G_4), y(G_4)$ and $z(G_4)$, where $G_4 \in [0, 6.74e - 5]$. On the other hand, the expression

$$\left(\frac{\beta_1 - \beta_2\beta_3}{\beta_4}\right),$$

has a magnitude of 10^{-1} for all values of $x(G_2), y(G_2)$ and $z(G_2)$, where $G_2 \in [0, 0.347]$. This indicates that the determining factor on the size of the determinant are the expressions

$$(x\lambda_{11} - z\lambda_{12}) \quad \text{and} \quad \left(\frac{z\lambda_{18}}{z + \lambda_{19}} - \lambda_{17} \right).$$

The expression $(x\lambda_{11} - z\lambda_{12})$ has an order of magnitude 10^4 , whereas

$$\left(\frac{z\lambda_{18}}{z + \lambda_{19}} - \lambda_{17} \right),$$

has an order of magnitude of 10^0 .

4.1 Sensitivity of the model to parameters

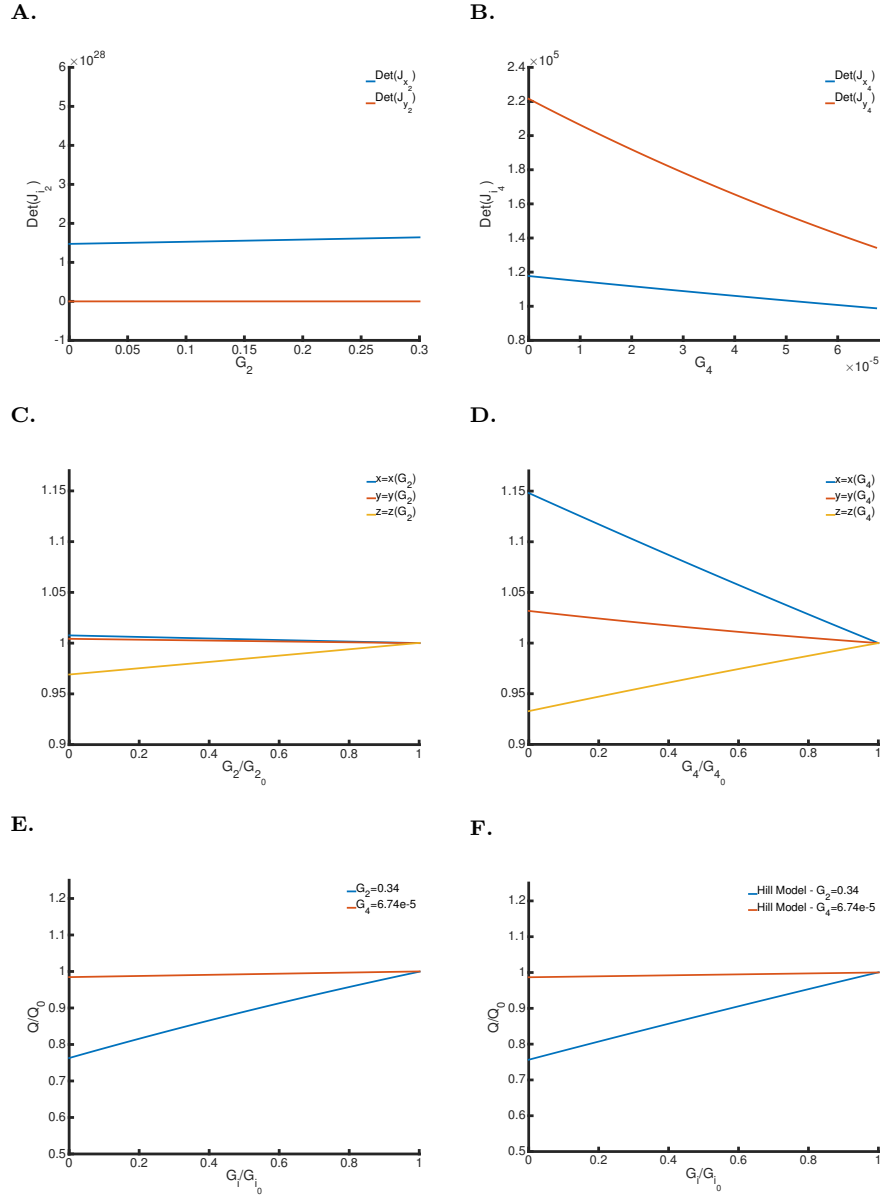


Fig. 1: **A.** Determinant of the Jacobian matrix of the system as a function of the parameter G_2 . **B.** Determinant of the Jacobian matrix of the system as a function of the parameter G_4 . **C.** Set of steady state values of the variables of the system as a function of G_2 . **D.** Set of steady state values of the variables of the system as a function of G_4 . **E.** Fluid flow steady states as a function of the parameter G_i ; where $i = 2$ or $i = 4$. **F.** Fluid flow steady states as a function of the parameter G_i ; where $i = 2$ or $i = 4$, this time with a Hill function-type model.

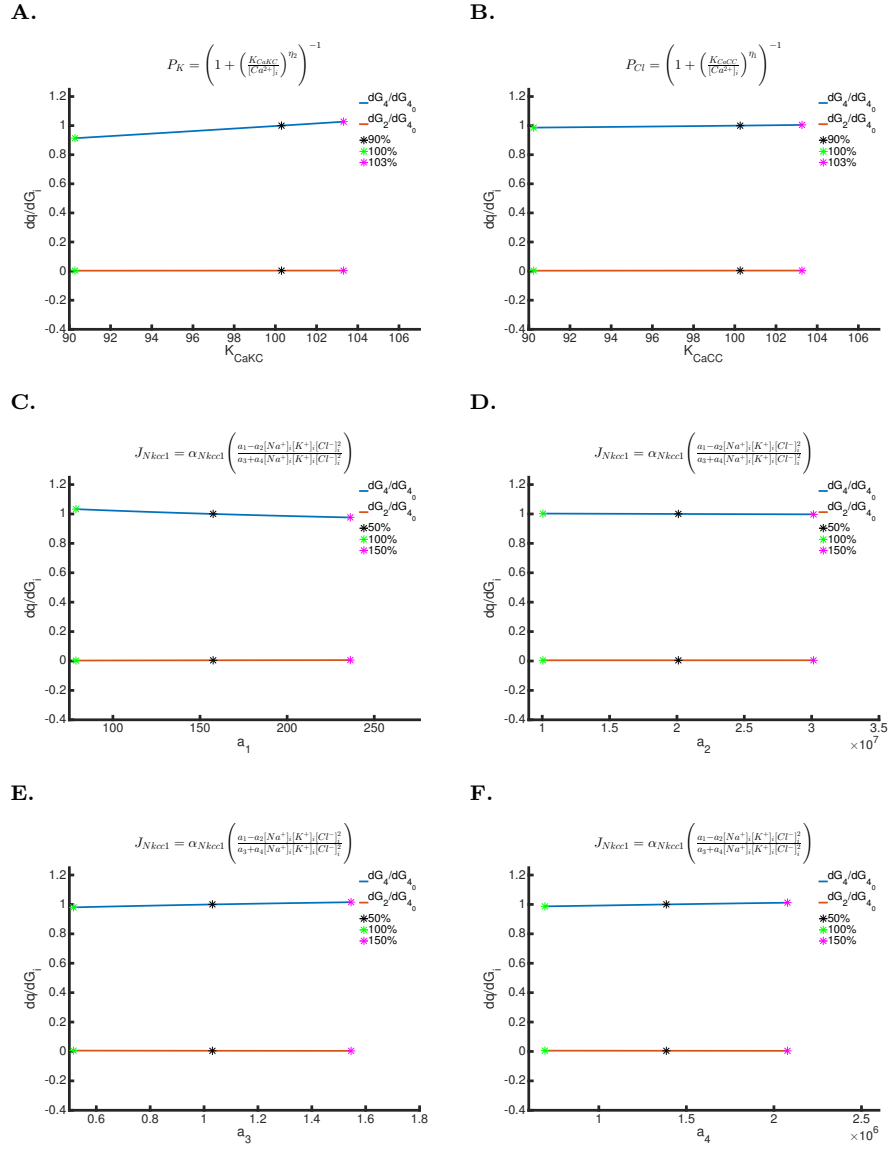


Fig. 2

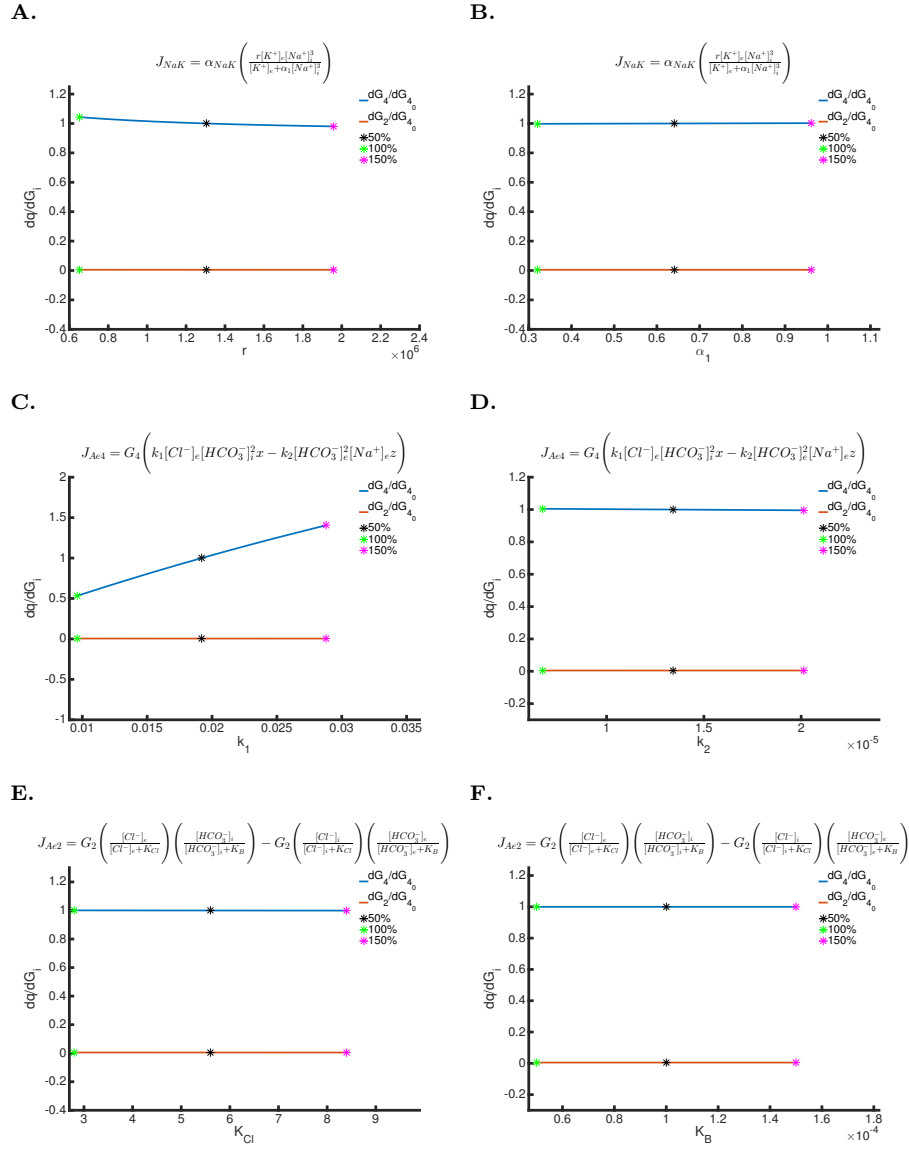
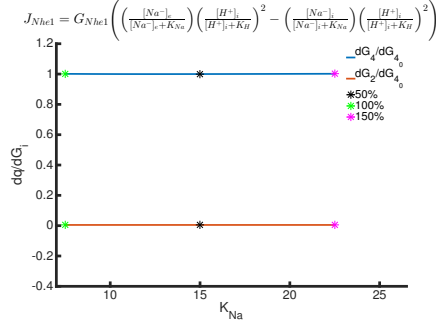


Fig. 3

A.



B.

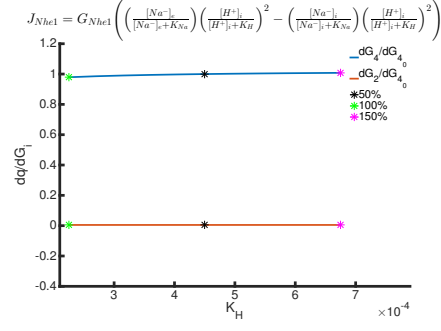


Fig. 4

Appendix

Nkcc1

The steady state flux through the Nkcc1 cotransporter is given by,

$$j_{\text{Nkcc1}} = \alpha_{\text{Nkcc1}} \left(\frac{a_1 - a_2 [\text{Na}^+]_i [\text{K}^+]_i [\text{Cl}^-]_i^2}{a_3 + a_4 [\text{Na}^+]_i [\text{K}^+]_i [\text{Cl}^-]_i^2} \right),$$

where α_{Nkcc1} is the cotransporter's density. Following the non-dimensionalisation steps in section ... we arrive at the non-dimensional flux,

$$j_{\text{Nkcc1}} = \left(\frac{\lambda_1 - \lambda_2 (xyz^2)}{\lambda_3 + \lambda_4 (xyz^2)} \right). \quad (13)$$

Table 2: Parameters of the model.

Parameter	Value
α_{Nkcc1}	7.21 (fmol / μm^3)
a_1	157.5 (s^{-1})
a_2	2.0096×10^7 (mM^4s^{-1})
a_3	1.0306 (s^{-1})
a_4	1.3852×10^6 (mM^4s^{-1})

$\text{Na}^+ \text{K}^+$ -ATPase

The steady state flux through the NaK-ATPase pump is given by,

$$j_{\text{NaK}} = \alpha_{\text{NaK}} \left(\frac{r [\text{K}^+]_e^2 [\text{Na}^+]_i^3}{[\text{K}^+]_e^2 + \alpha [\text{Na}^+]_i^3} \right),$$

where α_{NaK} is the cotransporter's density. Following the non-dimensionalisation steps in section ... we arrive at the non-dimensional flux,

$$j_{\text{NaK}} = \frac{\lambda_5 x^3}{\lambda_6 + \lambda_7 x^3}. \quad (14)$$

Table 3: Parameters of the model.

Parameter	Value
α_{NaK}	0.93 (fmol / μm^3)
r	1.305×10^6 (mM^3s^{-1})
α	0.641 (mM^{-1})

Ca²⁺-activated K⁺ channels (CaKC)

The open probability of the CaKC channels is given by,

$$P_{\text{CaKC}} = \left(\frac{[\text{Ca}^{2+}]_i}{[\text{Ca}^{2+}]_i + K_{\text{CaKC}}} \right)^{\eta_2},$$

where $\eta_2 = 2.54$ is the Hill coefficient and $K_{\text{CaKC}} = 0.182 \mu\text{M}$. Note that P_{CaKC} is already non-dimensional. The flux is defined as,

$$j_{\text{CaKC}} = \frac{G_{\text{CaKC}} P_{\text{CaKC}}}{F} (V_b - V_{\text{CaKC}}),$$

where V_{CaKC} is the Nernst potential defined by,

$$V_{\text{CaKC}} = \frac{RT}{Fz^K} \ln \left(\frac{[\text{K}^+]_e}{[\text{K}^+]_i} \right).$$

Here z^K is the ion's valence, T is the temperature in Kelvin, R the universal gas constant in SI units, and F Faraday's constant in SI units. Following the non-dimensionalisation procedure above we obtain,

$$j_{\text{CaKC}} = \lambda_8 - \lambda_9 \ln(y). \quad (15)$$

Ca²⁺-activated Cl⁻ channels (CaCC)

The open probability of the CaCC channels is given by,

$$P_{\text{CaCC}} = \left(\frac{[\text{Ca}^{2+}]_i}{[\text{Ca}^{2+}]_i + K_{\text{CaCC}}} \right)^{\eta_1},$$

where $\eta_1 = 1.46$ is the Hill coefficient and $K_{\text{CaCC}} = 0.26 \mu\text{M}$. Note that P_{CaCC} is already non-dimensional. The flux is defined as,

$$j_{\text{CaCC}} = \frac{G_{\text{CaCC}} P_{\text{CaCC}}}{F} (V_a + V_{\text{CaCC}}),$$

where V_{CaCC} is the Nernst potential defined by,

$$V_{\text{CaCC}} = \frac{RT}{Fz^{\text{Cl}}} \ln \left(\frac{[\text{Cl}^-]_l}{[\text{Cl}^-]_i} \right).$$

Here z^{Cl} is the ion's valence, T is the temperature in Kelvin, R the universal gas constant in SI units, and F Faraday's constant in SI units. Following the non-dimensionalisation procedure above we obtain,

$$j_{\text{CaCC}} = \lambda_{10} - \lambda_{11} \ln(z). \quad (16)$$

HCO₃⁻/Cl⁻ exchanger (Ae2)

The flux of the transporter is given by,

$$j_{\text{Ae2}} = G_2 \left[\left(\frac{[\text{Cl}^-]_e}{[\text{Cl}^-]_e + K_{\text{Cl}}} \right) \left(\frac{[\text{HCO}_3^-]_i}{[\text{HCO}_3^-]_i + K_{\text{B}}} \right) - \left(\frac{[\text{Cl}^-]_i}{[\text{Cl}^-]_i + K_{\text{Cl}}} \right) \left(\frac{[\text{HCO}_3^-]_e}{[\text{HCO}_3^-]_e + K_{\text{B}}} \right) \right].$$

Following the non-dimensionalisation presented above,

$$j_{\text{Ae2}} = \lambda_{12} - \frac{\lambda_{13}z}{z + \lambda_{14}}. \quad (17)$$

Table 4: Parameters of the model.

Parameter	Value
G_2	1.92 (amol / μm^3 s)
K_{Cl}	5.6 (mM)
K_{B}	10^4 (mM)

(Na^+) - $2\text{HCO}_3^-/\text{Cl}^-$ exchanger (Ae4)

The flux of the transporter is given by,

$$j_{\text{Ae4}} = G_4 \left(k_1 [\text{Cl}^-]_e [\text{HCO}_3^-]_i^2 [\text{Na}^+]_i - k_2 [\text{Cl}^-]_i [\text{HCO}_3^-]_e^2 [\text{Na}^+]_e \right)$$

Following the non-dimensionalisation presented above,

$$j_{\text{Ae4}} = \lambda_{15}x - \lambda_{16}z. \quad (18)$$

Table 5: Parameters of the model.

Parameter	Value
G_4	6.236 (amol / μm^3 s)
k_1	1.92×10^{-2} (s^{-1})
k_2	1.30×10^{-5} (s^{-1})

Na^+/H^+ exchanger (Nhe1)

The flux of the transporter is given by,

$$j_{\text{Nhe1}} = G_{\text{Nhe1}} \left[\left(\frac{[\text{Na}^+]_e}{[\text{Na}^+]_e + K_{\text{Na}}} \right) \left(\frac{[\text{H}^+]_i}{[\text{H}^+]_i + K_{\text{H}}} \right) - \left(\frac{[\text{Na}^+]_i}{[\text{Na}^+]_i + K_{\text{Na}}} \right) \left(\frac{[\text{H}^+]_e}{[\text{H}^+]_e + K_{\text{H}}} \right) \right].$$

Following the non-dimensionalisation presented above,

$$j_{\text{Nhe1}} = \lambda_{19} - \frac{\lambda_{20}x}{x + \lambda_{21}}. \quad (19)$$

Table 6: Parameters of the model.

Parameter	Value
G_1	6.6036 (amol / μm^3 s)
K_H	4.5×10^{-4} (mM)
K_{Na}	15 (mM)

Derivatives

We begin by calculating the derivatives:

$$\begin{aligned} \frac{df_1}{dx} = & -\frac{yz^2s^4(\lambda_1 - xyz^2s^4\lambda_2)\lambda_4}{(\lambda_3 + xyz^2s^4\lambda_4)^2} - \frac{yz^2s^4\lambda_2}{\lambda_3 + xyz^2s^4\lambda_4} + \frac{9x^5s^6\lambda_5\lambda_7}{(\lambda_6 + x^3s^3\lambda_7)^2} \\ & - \frac{9x^2s^3\lambda_5}{\lambda_6 + x^3s^3\lambda_7} + \frac{x\lambda_9}{(x + \lambda_{10})^2} - \frac{\lambda_9}{x + \lambda_{10}} - \lambda_{11}G_4, \\ \frac{df_1}{dy} = & -\frac{xz^2s^4(\lambda_2\lambda_3 + \lambda_1\lambda_4)}{(\lambda_3 + xyz^2s^4\lambda_4)^2}, \\ \frac{df_1}{dz} = & G_4\lambda_{12} - \frac{2\lambda_4s^4xyz(\lambda_1 - \lambda_2s^4xyz^2)}{(\lambda_3 + \lambda_4s^4xyz^2)^2} - \frac{2\lambda_2s^2xyz}{\lambda_3 + \lambda_4s^4xyz^2}. \end{aligned}$$

$$\frac{df_2}{dx} = \frac{6\lambda_5s^3x^2}{\lambda_7s^3x^3 + \lambda_6} - \frac{6\lambda_5\lambda_7s^6x^5}{(\lambda_7s^3x^3 + \lambda_6)^2} - \frac{\lambda_2s^4yz^2}{\lambda_4xys^4z^2 + \lambda_3} - \frac{\lambda_4s^4yz^2(-\lambda_2xys^4z^2 + \lambda_1)}{(\lambda_4xys^4z^2 + \lambda_3)^2},$$

$$\frac{df_2}{dy} = \frac{-\lambda_{14}}{y} - \frac{\lambda_2s^4xz^2}{\lambda_4xys^4z^2 + \lambda_3} - \frac{\lambda_4s^4xz^2(-\lambda_2xys^4z^2 + \lambda_1)}{(\lambda_4xys^4z^2 + \lambda_3)^2},$$

$$\frac{df_2}{dz} = -\frac{2\lambda_2s^4xyz}{\lambda_4xys^4z^2 + \lambda_3} - \frac{2\lambda_4s^4xyz(-\lambda_2xys^4z^2 + \lambda_1)}{(\lambda_4xys^4z^2 + \lambda_3)^2}.$$

$$\begin{aligned} \frac{df_3}{dx} = & \frac{\lambda_9x}{(\lambda_{10} + x)^2} - \frac{\lambda_9}{\lambda_{10} + x} - \frac{9\lambda_5s^3x^2}{\lambda_7s^3x^3 + \lambda_6} + \frac{9\lambda_5\lambda_7s^6x^5}{(\lambda_7s^3x^3 + \lambda_6)^2} \\ & - \frac{3\lambda_2s^4yz^2}{\lambda_4xys^4z^2 + \lambda_3} - \frac{3\lambda_4s^4yz^2(-\lambda_2xys^4z^2 + \lambda_1)}{(\lambda_4xys^4z^2 + \lambda_3)^2}, \end{aligned}$$

$$\frac{df_3}{dy} = \frac{-3\lambda_2s^4xz^2}{\lambda_4xys^4z^2 + \lambda_3} - \frac{3\lambda_4s^4xz^2(-\lambda_2xys^4z^2 + \lambda_1)}{(\lambda_4xys^4z^2 + \lambda_3)^2},$$

$$\frac{df_3}{dz} = -G_2\left(\frac{\lambda_{18}}{\lambda_{19} + z}\right) - \frac{\lambda_{18}z}{(\lambda_{19} + z)^2} - \frac{\lambda_{16}}{z} - \frac{6\lambda_2s^4xyz}{\lambda_4xys^4z^2 + \lambda_3} - \frac{6\lambda_4s^4xyz(-\lambda_2xys^4z^2 + \lambda_1)}{(\lambda_4xys^4z^2 + \lambda_3)^2}$$

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